

Torque Balance

The rotation of the wheel is governed by Newton's law $F = ma$, expressed as a balance between the applied torques and the rate of change of angular momentum. Let I denote the moment of inertia of the wheel. Note that in general I depends on t , because the distribution of water does. But this complication disappears if we wait long enough: as $t \rightarrow \infty$, one can show that $I(t) \rightarrow \text{constant}$ (Exercise 9.1.1). Hence, after the transients decay, the equation of motion is

$$I\dot{\omega} = \text{damping torque} + \text{gravitational torque}.$$

There are two sources of damping: viscous damping due to the heavy oil in the brake, and a more subtle “inertial” damping caused by a spin-up effect—the water enters the wheel at zero angular velocity but is spun up to angular velocity ω before it leaks out. Both of these effects produce torques proportional to ω , so we have

$$\text{damping torque} = -v\omega,$$

where $v > 0$. The negative sign means that the damping opposes the motion.

The gravitational torque is like that of an inverted pendulum, since water is pumped in at the top of wheel (Figure 9.1.6).

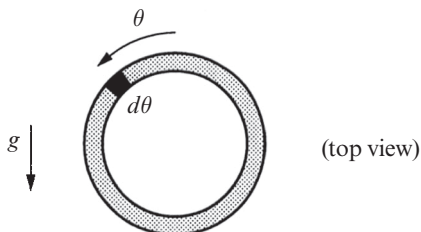
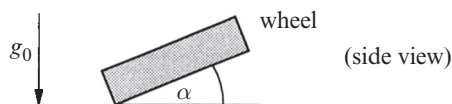


Figure 9.1.6

In an infinitesimal sector $d\theta$, the mass $dM = md\theta$. This mass element produces a torque

$$d\tau = (dM)gr \sin\theta = mgr \sin\theta d\theta.$$

To check that the sign is correct, observe that when $\sin\theta > 0$ the torque tends to *increase* ω , just as in an inverted pendulum. Here g is the effective gravitational



constant, given by $g = g_0 \sin \alpha$ where g_0 is the usual gravitational constant and α is the tilt of the wheel from horizontal (Figure 9.1.7). Integration over all mass elements yields

Figure 9.1.7

$$\text{gravitational torque} = gr \int_0^{2\pi} m(\theta, t) \sin \theta d\theta.$$

Putting it all together, we obtain the torque balance equation

$$I\dot{\omega} = -v\omega + gr \int_0^{2\pi} m(\theta, t) \sin \theta d\theta. \quad (3)$$

This is called an *integro-differential equation* because it involves both derivatives and integrals.

Amplitude Equations

Equations (2) and (3) completely specify the evolution of the system. Given the current values of $m(\theta, t)$ and $\omega(t)$, (2) tells us how to update m and (3) tells us how to update ω . So no further equations are needed.

If (2) and (3) truly describe the waterwheel's behavior, there must be some pretty complicated motions hidden in there. How can we extract them? The equations appear much more intimidating than anything we've studied so far.

A miracle occurs if we use Fourier analysis to rewrite the system. Watch!

Since $m(\theta, t)$ is periodic in θ , we can write it as a Fourier series

$$m(\theta, t) = \sum_{n=0}^{\infty} [a_n(t) \sin n\theta + b_n(t) \cos n\theta]. \quad (4)$$

By substituting this expression into (2) and (3), we'll obtain a set of **amplitude equations**, ordinary differential equations for the amplitudes a_n, b_n of the different *harmonics* or *modes*. But first we must also write the inflow as a Fourier series:

$$Q(\theta) = \sum_{n=0}^{\infty} q_n \cos n\theta. \quad (5)$$

There are no $\sin n\theta$ terms in the series because water is added *symmetrically* at the top of the wheel; the same inflow occurs at θ and $-\theta$. (In this respect, the waterwheel is unlike an ordinary, real-world waterwheel where asymmetry is used to drive the wheel in the same direction at all times.)

Substituting the series for m and Q into (2), we get

$$\begin{aligned} \frac{\partial}{\partial t} \left[\sum_{n=0}^{\infty} a_n(t) \sin n\theta + b_n(t) \cos n\theta \right] = & -\omega \frac{\partial}{\partial \theta} \left[\sum_{n=0}^{\infty} a_n(t) \sin n\theta + b_n(t) \cos n\theta \right] \\ & + \sum_{n=0}^{\infty} q_n \cos n\theta \\ & - K \left[\sum_{n=0}^{\infty} a_n(t) \sin n\theta + b_n(t) \cos n\theta \right]. \end{aligned}$$

Now carry out the differentiations on both sides, and collect terms. By orthogonality of the functions $\sin n\theta$, $\cos n\theta$, we can equate the coefficients of each harmonic separately. For instance, the coefficient of $\sin n\theta$ on the left-hand side is $\dot{a}_n$, and on the right it is $n\omega b_n - Ka_n$. Hence

$$\dot{a}_n = n\omega b_n - Ka_n. \quad (6)$$

Similarly, matching coefficients of $\cos n\theta$ yields

$$\dot{b}_n = -n\omega a_n - Kb_n + q_n. \quad (7)$$

Both (6) and (7) hold for all $n = 0, 1, \dots$

Next we rewrite (3) in terms of Fourier series. *Get ready for the miracle.* When we substitute (4) into (3), only one term survives in the integral, by orthogonality:

$$\begin{aligned} I\dot{\omega} &= -v\omega + gr \int_0^{2\pi} \left[\sum_{n=0}^{\infty} a_n(t) \sin n\theta + b_n(t) \cos n\theta \right] \sin \theta d\theta \\ &= -v\omega + gr \int_0^{2\pi} a_1 \sin^2 \theta d\theta \\ &= -v\omega + \pi gra_1. \end{aligned} \quad (8)$$

Hence, only a_1 enters the differential equation for $\dot{\omega}$. But then (6) and (7) imply that a_1, b_1 , and ω form a closed system—these three variables are decoupled from all the other $a_n, b_n, n \neq 1$! The resulting equations are

$$\begin{aligned} \dot{a}_1 &= \omega b_1 - Ka_1 \\ \dot{b}_1 &= -\omega a_1 - Kb_1 + q_1 \\ \dot{\omega} &= (-v\omega + \pi gra_1)/I. \end{aligned} \quad (9)$$

(If you're curious about the higher modes $a_n, b_n, n \neq 1$, see Exercise 9.1.2.)

We've simplified our problem tremendously: the original pair of integro-partial differential equations (2), (3) has boiled down to the three-dimensional system (9). It turns out that (9) is equivalent to the Lorenz equations! (See Exercise 9.1.3.) Before we turn to that more famous system, let's try to understand a little about (9). No one has ever *fully* understood it—its behavior is fantastically complex—but we can say something.

Fixed Points

We begin by finding the fixed points of (9). For notational convenience, the usual asterisks will be omitted in the intermediate steps.

Setting all the derivatives equal to zero yields

$$a_1 = \omega b_1 / K \quad (10)$$

$$\omega a_1 = q_1 - Kb_1 \quad (11)$$

$$a_1 = v\omega/\pi gr. \quad (12)$$

Now solve for b_1 by eliminating a_1 from (10) and (11):

$$b_1 = \frac{Kq_1}{\omega^2 + K^2}.$$

Equating (10) and (12) yields $\omega b_1/K = v\omega/\pi gr$. Hence $\omega = 0$ or

$$b_1 = Kv/\pi gr. \quad (14)$$

Thus, there are two kinds of fixed point to consider:

1. If $\omega = 0$, then $a_1 = 0$ and $b_1 = q_1/K$. This fixed point

$$(a_1^*, b_1^*, \omega^*) = (0, q_1/K, 0) \quad (15)$$

corresponds to a state of *no rotation*; the wheel is at rest, with inflow balanced by leakage. We're not saying that this state is stable, just that it exists; stability calculations will come later.

2. If $\omega \neq 0$, then (13) and (14) imply $b_1 = Kq_1/(\omega^2 + K^2) = Kv/\pi gr$. Since $K \neq 0$, we get $q_1/(\omega^2 + K^2) = v/\pi gr$. Hence

$$(\omega^*)^2 = \frac{\pi gr q_1}{v} - K^2. \quad (16)$$

If the right-hand side of (16) is positive, there are two solutions, $\pm\omega^*$, corresponding to *steady rotation* in either direction. These solutions exist if and only if

$$\frac{\pi gr q_1}{K^2 v} > 1. \quad (17)$$

The dimensionless group in (17) is called the **Rayleigh number**. It measures how hard we're driving the system, relative to the dissipation. More precisely, the ratio in (17) expresses a competition between g and q_1 (gravity and inflow, which tend to spin the wheel), and K and v (leakage and damping, which tend to stop the wheel). So it makes sense that steady rotation is possible only if the Rayleigh number is large enough.

The Rayleigh number appears in other parts of fluid mechanics, notably convection, in which a layer of fluid is heated from below. There it is proportional to the difference in temperature from bottom to top. For small temperature gradients, heat is conducted vertically but the fluid remains motionless. When the Rayleigh number increases past a critical value, an instability occurs—the hot fluid is less

dense and begins to rise, while the cold fluid on top begins to sink. This sets up a pattern of convection rolls, completely analogous to the steady rotation of our waterwheel. With further increases of the Rayleigh number, the rolls become wavy and eventually chaotic.

The analogy to the waterwheel breaks down at still higher Rayleigh numbers, when turbulence develops and the convective motion becomes complex in space as well as time (Drazin and Reid 1981, Bergé et al. 1984, Manneville 1990). In contrast, the waterwheel settles into a pendulum-like pattern of reversals, turning once to the left, then back to the right, and so on indefinitely (see Example 9.5.2).

9.2 Simple Properties of the Lorenz Equations

In this section we'll follow in Lorenz's footsteps. He took the analysis as far as possible using standard techniques, but at a certain stage he found himself confronted with what seemed like a paradox. One by one he had eliminated all the known possibilities for the long-term behavior of his system: he showed that in a certain range of parameters, there could be no stable fixed points and no stable limit cycles, yet he also proved that all trajectories remain confined to a bounded region and are eventually attracted to a set of zero volume. What could that set be? And how do the trajectories move on it? As we'll see in the next section, that set is the strange attractor, and the motion on it is chaotic.

But first we want to see how Lorenz ruled out the more traditional possibilities. As Sherlock Holmes said in *The Sign of Four*, "When you have eliminated the impossible, whatever remains, however improbable, must be the truth."

The Lorenz equations are

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz.\end{aligned}\tag{1}$$

Here $\sigma, r, b > 0$ are parameters: σ is the *Prandtl number*, r is the Rayleigh number, and b has no name. (In the convection problem it is related to the aspect ratio of the rolls.)

Nonlinearity

The system (1) has only two nonlinearities, the quadratic terms xy and xz . This should remind you of the waterwheel equations (9.1.9), which had two nonlinearities, ωa_1 and ωb_1 . See Exercise 9.1.3 for the change of variables that transforms the waterwheel equations into the Lorenz equations.

Symmetry

There is an important *symmetry* in the Lorenz equations. If we replace $(x, y) \rightarrow (-x, -y)$ in (1), the equations stay the same. Hence, if $(x(t), y(t), z(t))$ is a solution, so is $(-x(t), -y(t), z(t))$. In other words, all solutions are either symmetric themselves, or have a symmetric partner.

Volume Contraction

The Lorenz system is *dissipative*: volumes in phase space contract under the flow. To see this, we must first ask: how do volumes evolve?

Let's answer the question in general, for any three-dimensional system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. Pick an arbitrary closed surface $S(t)$ of volume $V(t)$ in phase space. Think of the points on S as initial conditions for trajectories, and let them evolve for an infinitesimal time dt . Then S evolves into a new surface $S(t + dt)$; what is its volume $V(t + dt)$?

Figure 9.2.1 shows a side view of the volume.

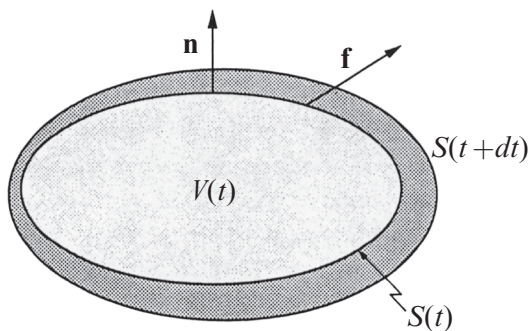


Figure 9.2.1

Let $\mathbf{n}$ denote the outward normal on S . Since $\mathbf{f}$ is the instantaneous velocity of the points, $\mathbf{f} \cdot \mathbf{n}$ is the outward normal component of velocity. Therefore in time dt a patch of area dA sweeps out a volume $(\mathbf{f} \cdot \mathbf{n} dt)dA$, as shown in Figure 9.2.2.

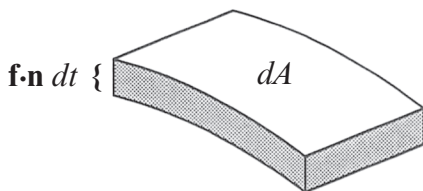


Figure 9.2.2

Hence

$$V(t + dt) = V(t) + (\text{volume swept out by tiny patches of surface, integrated over all patches}),$$

so we obtain

$$V(t + dt) = V(t) + \int_S (\mathbf{f} \cdot \mathbf{n} dt) dA.$$

Hence

$$\dot{V} = \frac{V(t + dt) - V(t)}{dt} = \int_S \mathbf{f} \cdot \mathbf{n} dA.$$

Finally, we rewrite the integral above by the divergence theorem, and get

$$\dot{V} = \int_V \nabla \cdot \mathbf{f} dV. \quad (2)$$

For the Lorenz system,

$$\begin{aligned} \nabla \cdot \mathbf{f} &= \frac{\partial}{\partial x} [\sigma(y - x)] + \frac{\partial}{\partial y} [rx - y - xz] + \frac{\partial}{\partial z} [xy - bz] \\ &= -\sigma - 1 - b < 0. \end{aligned}$$

Since the divergence is constant, (2) reduces to $\dot{V} = -(\sigma + 1 + b)V$, which has solution $V(t) = V(0)e^{-(\sigma + 1 + b)t}$. Thus *volumes in phase space shrink exponentially fast*.

Hence, if we start with an enormous solid blob of initial conditions, it eventually shrinks to a limiting set of zero volume, like a balloon with the air being sucked out of it. All trajectories starting in the blob end up somewhere in this limiting set; later we'll see it consists of fixed points, limit cycles, or for some parameter values, a strange attractor.

Volume contraction imposes strong constraints on the possible solutions of the Lorenz equations, as illustrated by the next two examples.

EXAMPLE 9.2.1:

Show that there are no quasiperiodic solutions of the Lorenz equations.

Solution: We give a proof by contradiction. If there were a quasiperiodic solution, it would have to lie on the surface of a torus, as discussed in Section 8.6, and this torus would be *invariant* under the flow. Hence the volume inside the torus would be constant in time. But this contradicts the fact that all volumes shrink exponentially fast. ■